Ze-Yu Zhong

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Summary

Research Fellow in Econometrics at the University of Melbourne, focusing on high-dimensional quantile models for macroeconomic and financial data during extreme events. I completed my PhD in Econometrics in 2025 at Monash University, where I developed new methods for analyzing structural breaks, forecasting during periods of instability, and high-dimensional structural analysis for macroeconomic factor models. Extensive experience in developing econometric theory, models, and algorithms, as well as applying these methods to real-world economic and financial datasets.

Research Experience

Research Fellow University of Melbourne, September 2025 - Present

- Developed new econometric methods for high-dimensional quantile regression models
- Designed R package for estimation and inference for high-dimensional time series models
  - Methods include Ridge, LASSO, Elastic Net, and post-selection inference methods such as debiasing, for both mean and quantile regression models
  - Implemented high-dimensional family-wise inference methods to ensure valid network discovery
  - Visualised contagion networks of global financial market of extreme events

PhD Candidate Monash University, March 2021 - April 2025

- *"Dynamic Factor Models: Structural Breaks, Forecasting, and Structural Analysis"*
- Developed new econometric methods for factors models, including structural breaks, forecasting with structural breaks, and structural analysis using external instruments

Research Assistant November 2019 - June 2025

- Research assistant for Donald Poskitt and Xueyan Zhao
  - *Hausdorff Confidence Regions for Average Treatment Effect Intersection Bounds*
  - Constructed large Monte Carlo designs to investigate performance of QMLE, Half-median Unbiased Estimation, and Bootstrap based partial identification of Treatment Effect Bounds, with application to Australia Healthcare data
  - Converted R code to C++ and High-performance Computing for 100x speedup
- Research assistant for Bonsoo Koo
  - Implemented bootstrap based inference for non-parametric estimation of structural via an object-oriented programming approach with S3
- Research assistant for Benjamin Wong
  - Queried large panels of price indices from U.S. Bureau of Labor Statistics and U.S. Bureau of Economic Analysis, using API calls
- Research assistant for David Frazier
  - Investigated and implemented performance of DeepAR forecasting algorithms using Amazon AWS, including EC2 and Sagemaker for use in forecasting stock returns

Research Projects

Network Discovery for High-dimensional Quantile Vector-Autoregressive Models (Current)

- Developed penalised quantile regression methods for high-dimensional time series data, to study contagion networks of extreme macroeconomic events

Identification and Estimation of Structural Factor Models with External Instruments (solo)

- Developed a structural model which combined large macroeconomic datasets and external high-frequency signals to identify and estimate effects of monetary policy
- Implemented algorithms to automatically combine and select external signals, leading a 50% improvement in efficiency
- Presentations:
  - ANZESG 2024 (Melbourne), recipient of Best Speaker Prize (AUD 500)

Factor Augmented Forecasting Subject to Structural Breaks in the Factor Structure

- Collaborated internationally to improve estimation of latent factors by purging out effects of macroeconomic instability, leading to a 3% improvement over prevailing benchmarks for large scale macroeconomic forecasting when backtested
- Combined forecasts together using cross-validation based ensemble learning to produce predictions robust to periods of instability, such as COVID-19 Pandemic
- Presentations:
  - International Association for Applied Econometrics Conference (Xiamen and Thessaloniki, 2024), International Symposium on Forecasting (Dijon, 2024), ESAM 2024 (Melbourne)

Disentangling Structural Breaks in Factor Models for Macroeconomic Data

- Published at Journal of Business & Economic Statistics (2025)
- Decomposed sources of structural change in high dimensional factor models for macroeconomic data, quantifying a 60% decrease in macroeconomic volatility since the 1980s
- Developed statistical procedures to test and quantify changes in common and idiosyncratic sources of volatility
- Presentations:
  - Continuing Education in Macroeconometrics Workshop (Melbourne, 2022), City University of Hong Kong Seminar (2022), BI Norwegian Business School Seminar (Oslo, 2022), EC^2 (Paris, 2022), University of Geneva Brown Bag Seminar (2022), ESAM (Sydney, 2023), International Panel Data Conference (Orleans, 2024)

Education

2021 - 2025 Doctor of Philosophy (Econometrics) at **Monash University**
"Dynamic Factor Models: Structural Breaks, Forecasting, and Structural Analysis"

- Recipient of Post-Publication Award to support write up and publication (AUD 6,000)
- Recipient of Student Travel Grant for International Symposium on Forecasting (USD 3,500)
- Recipient of Monash University Staff Grant (AUD 12,000)

2019 - 2019 Bachelor of Commerce (Honours in Econometrics) at **Monash University** (79 WAM)
"Evaluation of Machine Learning in Empirical Finance"

2016 - 2018 Bachelor of Actuarial Science at **Monash University** (80 WAM)

Teaching Experience

Teaching Associate February 2020 - June 2025

- ETC1000 Business Statistics
  - Summary Statistics, Pivot Tables, Hypothesis Testing, (Microsoft Excel)
- ETC2410/ETC3440 Introductory Econometrics
  - Ordinary Least Squares Regression, Hypothesis Testing, Introduction to Time Series (EViews)
- ETC3410/ETC5341/BEX3410 Applied Econometrics
  - Binary Outcome Models, Instrumental Variables Regression, Panel Data Models (R/RStudio)
  - Recipient of Purple Letter for recognition of excellence in teaching
- ETC2420/ETC5242 Statistical Thinking
  - Non-parametric/parametric inference, Data Visualisation, Bayesian Inference (R/RStudio)
- ETX2250 Data Visualisation and Analytics
  - Data wrangling, Data Visualisation, SQL queries, supervised and non-supervised machine learning
- ETC4420/5420 Microeconometrics
  - Multinomial Outcome Models, Truncated and Censored Regression, Sample Selection Models (STATA)

Referee Service

Studies in Nonlinear Dynamics & Econometrics

References

Bonsoo Koo	Associate Professor, Monash University	bonsoo.koo@monash.edu
Xu Han	Associate Professor, City University of Hong Kong	xuhan25@cityu.edu.hk
Donald Poskitt	Emeritus Professor, Monash University	donald.poskitt@monash.edu